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Group Project
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PMFBY CLAIM TRIAGE INTELLIGENCE

Human-Supervised Verification Prioritisation using Agricultural Evidence

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Executive Summary

PMFBY Claim Triage Intelligence is an end-to-end Big Data Analytics project designed to improve the order in which crop-loss cases are taken up for human verification. The project integrates PMFBY/DES agricultural and insurance context, district references, NASA POWER weather data and MODIS vegetation evidence on Databricks, then expands the real aggregate context into a two-million-record synthetic workload for distributed modelling and policy testing.

The final release has 4,497 district-season-year contexts and generates exactly 2,000,000 synthetic claim scenarios. The pipeline preserves missing-data provenance, separates observable model inputs from hidden simulation truth, uses versioned Delta outputs and a fixed seed, and passes ten runtime acceptance checks. Logistic Regression and Random Forest classifiers are evaluated using a temporal train-validation-test design.

The central managerial finding is not that a complex model dominates. At 20% priority-review capacity, the full-feature Random Forest captures 27.97% of simulated loss cases, while a transparent weather-vegetation rule captures 27.78% and random allocation captures 19.94%. The Random Forest therefore adds only 0.19 percentage points over the transparent rule. The recommended decision is to use the simple evidence rule as the first supervised baseline and require measured operational benefit before introducing additional model complexity.

Decision in one sentence: Use the integrated evidence pipeline to rank cases for human verification, begin with the transparent weather-vegetation rule, and adopt more complex ML only if future real-world validation shows a meaningful incremental benefit.

Project Objective and Business Question

The original proposal considered automatic claim approval, rejection and manual review. During development, that objective was deliberately narrowed to a more realistic decision problem: which cases should be assessed first when verification capacity is limited? The system does not determine entitlement, payment or fraud. It produces a ranked worklist so that cases showing stronger evidence of loss can enter the first-review queue while all remaining cases continue through standard human review.

- **Business question:** How can heterogeneous agricultural, weather and vegetation evidence be combined at scale to prioritise human verification?
- **Engineering question:** Can a reproducible distributed pipeline retain incomplete contexts without disguising estimated values as observations?
- **Decision question:** Does machine learning add enough operational value over an inspectable evidence rule to justify additional complexity?

Business Context and Data Availability

The project addresses a real data constraint. Publicly accessible farmer-level PMFBY claim records with verified field-level crop-loss outcomes are not available for the proposed modelling task. A two-million-record synthetic workload was therefore constructed to provide the scale required for demonstrating Big Data ingestion, storage, distributed processing, feature engineering and machine learning.

The synthetic layer is a controlled experimental workload attached to real district-season-year evidence, not two million independent farmer observations. Its label mechanism deliberately uses observable environmental stress signals - rainfall anomalies, vegetation anomalies, temperature and heavy-rain events - together with an independent hidden shock so that outcomes are stochastic rather than deterministic. Model performance therefore measures how effectively the pipeline learns and operationalises the declared simulated relationship; it is not an empirical discovery of causal PMFBY loss drivers.

Analytical Approach Selection. The supervised target is binary simulated loss versus no loss, making classification the appropriate primary model family. Logistic Regression and Random Forest provide two distinct PySpark MLlib classifiers. Regression would require a continuous target, while clustering would answer a separate segmentation question - for example, identifying district risk zones - and is better treated as a future managerial extension rather than inserted into the core claim-prioritisation pipeline.

Data Understanding: The 5 Vs

Dimension	Evidence and Interpretation
Volume	4,497 master contexts expand to 2,000,000 synthetic scenarios for distributed processing and modelling.
Velocity	Seasonal and annual batch processing. Same-season summaries make the current experiment retrospective rather than real-time.
Variety	PMFBY/DES context, district reference and shapefile data, NASA POWER weather, and MODIS NDVI/EVI vegetation evidence.
Veracity	Observed, recovered and estimated values remain distinguishable. 1,865 contexts are observed/recovered complete, 1,495 use limited-history evidence and 1,137 use regional fallback.
Value	The integrated evidence supports comparable priority-review queues at fixed capacity and shows how scarce first-review slots can be concentrated on higher-risk cases.

The useful part of this dataset is that the sources can be linked and traced. PMFBY/DES provide the insurance and agricultural context, NASA POWER supplies weather information, and MODIS adds NDVI/EVI vegetation measures. Together, they provide evidence for comparing district-season-year contexts. They cannot confirm what happened on a particular farmer's field. Generating more scenarios does not remove that limitation.

Data Sources and Analytical Role

Source	Role in the Project
PMFBY / DES	Insurance and crop context at aggregate geographic and seasonal levels.
District references / shapefile	Canonical district identities and spatial matching across sources.
NASA POWER	Daily meteorological observations used for temperature, rainfall, completeness and anomaly features.
MODIS vegetation indices	NDVI and EVI evidence used to represent vegetation condition and prior-year deviation.
Synthetic scenario layer	Claim-level synthetic workload used for distributed supervised experiments; labels remain explicitly synthetic.

Enterprise Architecture and Big Data Pipeline

The pipeline runs on Databricks. PySpark joins the sources, prepares seasonal summaries, generates the synthetic cases and fits the models. Intermediate and final results are saved as versioned Delta tables, and MLflow records the model artifacts. This lets the analysis be followed from the integrated data through to the review queues.

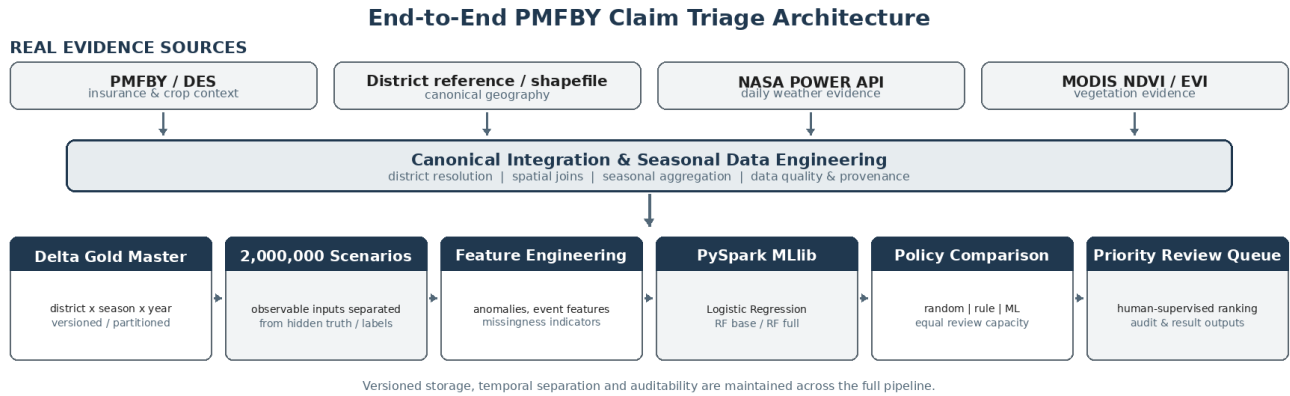


Figure 1: End-to-end architecture from heterogeneous evidence sources to an auditable priority-review queue.

Architecture Choices

Design Area	Implementation
Distributed processing	PySpark on Databricks for joins, transformations, scenario expansion and MLlib training.
Storage	Delta Lake tables for persistent, versioned intermediate and final layers.
Data organisation	District-season-year Gold layer; year/season partitioning where specified by publishing logic.
Reproducibility	Fixed seed, versioned run namespace and source hashes identify the executed experiment.
Auditability	Observable inputs, hidden simulator information, labels, predictions and review decisions are kept in separate tables.

HDFS, Sqoop and Kafka are not claimed as implemented components because the final workload is a batch analytics pipeline rather than a streaming system. The implemented evidence for Big Data capability is distributed execution, versioned storage, reproducible intermediate layers and a two-million-record analytical workload, rather than an unsupported claim that a single machine could not process the data. The excerpt below shows the fail-if-present, run-namespaced Delta publication pattern used to preserve reproducibility.

Implementation excerpt 1: Versioned Delta publishing logic (abridged from the executed notebook)

```
destination = table(name)
assert destination.startswith(PREFIX) and "_v2_2_" in destination
view = "pmfby_stage_" + RUN_ID + "_" + name
df.createOrReplaceTempView(view)
spark.sql("CREATE TABLE " + destination + " USING DELTA"
          + partition_sql + " AS SELECT * FROM " + view)
result = spark.table(destination)
```

Data Engineering and Feature Preparation

Canonical Integration and Context Retention

District identities are normalised before cross-source joins. Historical state and district references are used to recover exact matches without silently inventing geography. The final repair recovers 256 context-level vegetation matches and 18 centroid mappings, allowing all 4,497 contexts to enter scenario generation.

Missing Values and Provenance

Incomplete evidence is retained transparently instead of being removed by complete-case filtering. For non-anomaly features, the pipeline uses an observation or exact recovery where available, followed by state-year-season and then year-season fallback medians. Missing anomaly values use a neutral model placeholder together with an explicit missingness indicator. Separate fields preserve the model value, missing flag and fill method. The implementation excerpt below shows how model values, missingness flags and provenance labels are stored separately.

Implementation excerpt 2: Missingness provenance for anomaly features

```
for c in ["ndvi_anomaly", "weather_precip_anomaly_pct"]:
    gold = (gold
            .withColumn(c + "_model",
                        F.coalesce(F.col(c), F.lit(0.)).cast("double"))
            .withColumn(c + "_missing", F.col(c).isNull().cast("double"))
            .withColumn(c + "_fill_method",
                        F.when(F.col(c).isNotNull(), "observed_prior_year_anomaly")
                        .otherwise("no_prior_baseline_neutral_placeholder")))
```

Feature Engineering

The full model uses 16 numeric features and their missingness indicators. The base set covers premium rate, crop count, temperature, rainfall anomaly, NDVI, EVI, NDVI anomaly and weather completeness. The additional features describe dry periods, heat, heavy rain and rainfall extremes. The two Random Forest versions use these different feature sets so that their queue performance can be compared.

Engineering Decision	Purpose
Geographic repair	Recover exact matches first; avoid fuzzy district joins that could silently assign the wrong geography.
Missingness	Retain incomplete contexts with model placeholders plus explicit *_missing and *_fill_method provenance.
Anomalies	Use prior-history rainfall and NDVI deviations without later-year baselines.
Scaling	StandardScaler is included in the shared pipeline; useful for Logistic Regression, neutral for the tree model.
Temporal leakage control	Training, threshold selection and final testing are separated by year.

Distributed Machine Learning Implementation

Two classification approaches are implemented in PySpark MLlib: Logistic Regression and Random Forest. Random Forest is estimated with both a base and a full feature specification. The binary label means simulated loss occurred or did not occur; it is not a historical approval/rejection outcome. The shared pipeline below makes the feature assembly, scaling and estimator settings explicit.

Component	Executed Setting
Training period	2018-2020; deterministic 20% sample = 254,883 scenarios
Validation	2021; used for threshold selection

Component	Executed Setting
Test	2022; all 366,591 scenarios, including 96,740 simulated loss cases
Logistic Regression	maxIter = 60; regParam = 0.01
Random Forest	60 trees; maxDepth = 8; fixed seed = 20260907
Pipeline	VectorAssembler -> StandardScaler -> estimator

Implementation excerpt 3: Shared PySpark MLlib feature and estimator pipeline

```

assembler = VectorAssembler(
    inputCols=feature_names, outputCol="assembled", handleInvalid="error")
scaler = StandardScaler(
    inputCol="assembled", outputCol="features", withMean=False, withStd=True)

estimator = (
    LogisticRegression(maxIter=60, regParam=0.01, labelCol="label")
    if kind == "lr" else
    RandomForestClassifier(numTrees=60, maxDepth=8,
                          seed=SEED, labelCol="label")
)

fitted = Pipeline(stages=[assembler, scaler, estimator]).fit(
    train.select(*(feature_names + ["label"])))

```

Observable scenarios, hidden simulation truth and labels are physically separated. Rankings are persisted before labels are joined for evaluation. This prevents direct access to hidden simulator information during inference and makes the final experiment substantially more defensible than the earlier policy-label formulation.

Context-level granularity check: A saved-output evaluation of all 843 test-year district-season-year contexts found that Logistic Regression and both Random Forest specifications assign identical scores to synthetic cases within the same context. The current system therefore prioritises contextual risk expressed through a claim workload, rather than differentiating individual farmers within that context. Claim IDs are used only as deterministic tie-breakers among equal-scored cases; farmer-level differentiation would require field- or claim-specific predictors that are not available in the source data.

Model Evaluation and Interpretation

Model	ROC AUC	PR AUC	Brier Score
Logistic Regression	0.5653	0.3101	0.1938
Random Forest - full	0.5719	0.3266	0.1913
Random Forest - base	0.5670	0.3267	0.1915

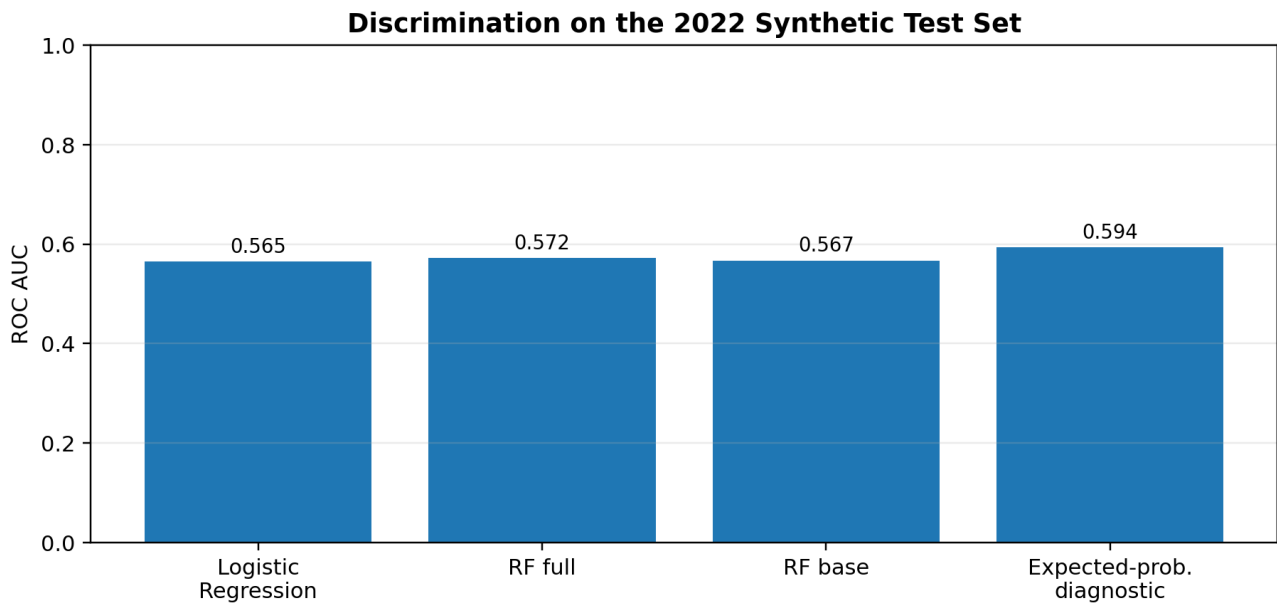


Figure 2: ROC AUC comparison on the 2022 synthetic test set. The expected-probability diagnostic (0.594) is shown only as a simulation reference, not as an operational model.

The models show modest discrimination because the simulated loss mechanism deliberately includes irreducible randomness. The full-feature Random Forest reaches ROC AUC 0.5719, close to the expected-probability diagnostic of 0.594 under the declared simulator. Model evaluation therefore focuses on whether rankings improve the actual review-capacity decision rather than treating AUC alone as the business objective.

Threshold diagnostic: The classification thresholds did not work well on the test set: every 2022 case was classified as a loss. This happened with the cutoffs selected using validation F1. The scores still varied across contexts, however, so they could be used to order cases. The queue comparison therefore evaluates the highest-ranked 10%, 20% and 30% at the same review capacity for each strategy. This tests ranking performance; it does not turn the failed yes/no classification into a successful result.

Equal-Capacity Policy Comparison

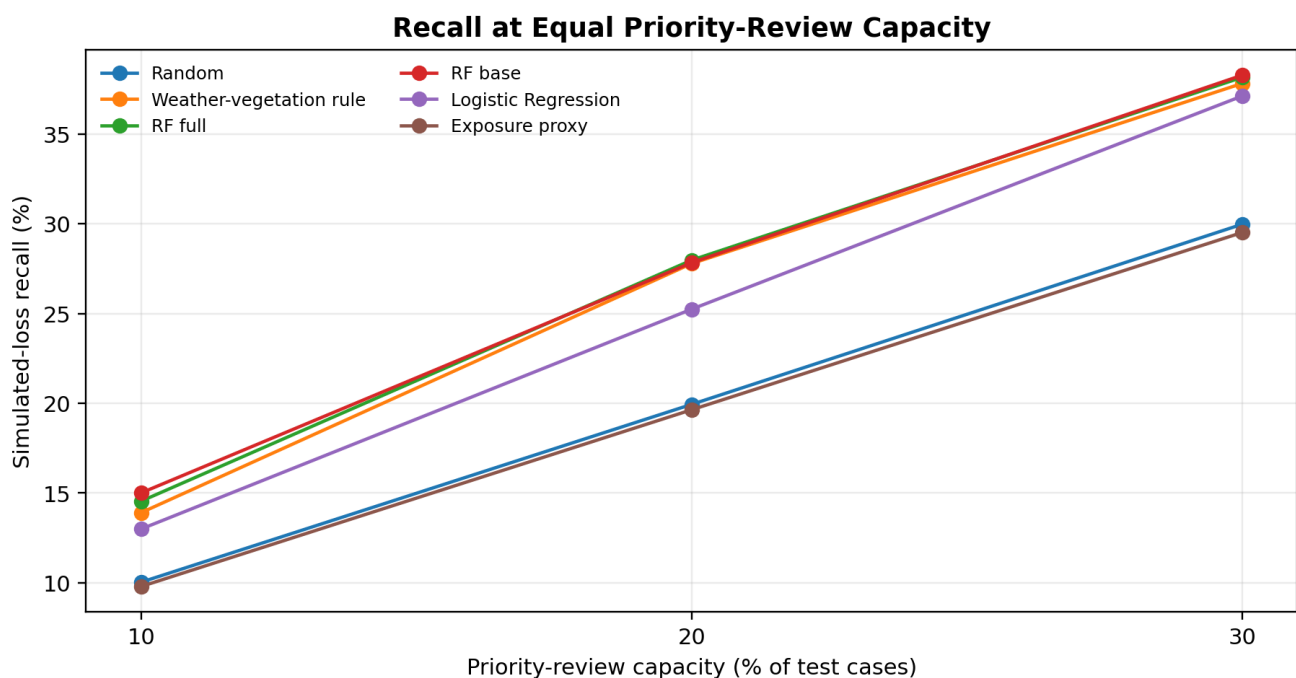


Figure 3: Simulated-loss recall at 10%, 20% and 30% priority-review capacity.

Strategy	10% Recall	20% Recall	30% Recall
Random	10.04%	19.94%	29.97%
Weather-vegetation rule	13.91%	27.78%	37.81%
RF full	14.55%	27.97%	38.15%
RF base	15.01%	27.83%	38.27%
Logistic Regression	13.00%	25.24%	37.11%
Exposure proxy	9.80%	19.64%	29.51%

Each strategy had 73,318 first-review slots at 20% capacity. Random selection placed 19,293 simulated loss cases in that queue. The weather-vegetation rule placed 26,876, an increase of 7,583 cases, or about 39.3% relative to random selection. Random Forest placed 27,058 loss cases in the queue, adding only 182 over the rule. These counts describe the simulation, not verified farmer outcomes.

Business interpretation: Within this simulation, the weather-vegetation rule captures more loss cases than random selection at the same capacity. Random Forest adds little beyond the rule. The results do not support a claim that ML offers a large additional benefit.

Uncertainty diagnostic: The saved-output check also examined how sensitive the RF-rule difference was to the mix of contexts. It resampled whole contexts 2,000 times, keeping the existing scores and queue membership fixed. No models were refitted and no queues were reranked. The observed recall difference was +0.188 percentage points; the conditional diagnostic interval ran from -1.506 to +1.996 points. Because this range includes zero, the check does not establish a clear RF advantage. It also does not prove that the two methods are equivalent or predict how they would perform on actual claims.

Business Insights

- **Evidence beats random sequencing.** At 20% capacity, the rule captures 27.78% of simulated losses, compared with 19.94% under random selection. Both queues contain the same number of cases.
- **Complexity does not automatically create value.** The full Random Forest leads the rule by only 0.19 percentage points at that capacity. It is not consistently the best model across budgets: the base-feature forest has higher recall at 10% and 30%.
- **Feature count is not a decision criterion.** The extra event features do not consistently improve queue performance. Their inclusion should be judged by the result they produce, rather than by the number of inputs added.
- **The output is a worklist, not an automated entitlement engine.** The intended output is an ordered worklist for human assessment. Cases outside the priority queue remain in standard review; the model does not decide entitlement.
- **The project creates a measurable baseline for future validation.** The saved outputs provide a reference for later testing. If operational evidence becomes available, the same comparison can be extended to assessor effort, resolution time, appeals and verified loss outcomes. None of those operational benefits has been measured here.

Strategic Recommendations

Recommendation 1: Evaluate the Transparent Rule in Shadow Mode

Use the weather-vegetation rule in shadow mode alongside the existing assessment process. It delivers near-equal recall to Random Forest while being easier to inspect, explain and maintain. No binding claim decision should be generated from the current synthetic labels.

Recommendation 2: Begin with the Tested 20% Priority Allocation

Use 20% of cases as a declared starting priority queue for the supervised trial, with the remaining cases retained in standard review. This is a tested middle setting rather than an optimised staffing recommendation; actual assessor-hours and case complexity should determine any future capacity choice.

Recommendation 3: Collect the Right Real-World Target

When verified outcomes become accessible, prioritise field-level evidence of crop loss such as yield shortfall, Crop Cutting Experiment outcomes or technology-based yield estimates. Historical approval/rejection decisions should not automatically become the target because they may reflect paperwork, administrative rules and earlier review bias in addition to crop loss.

Recommendation 4: Require ML to Earn Its Complexity

Compare any future ML model against the transparent rule at equal operational capacity. Adopt the more complex approach only if it produces a material, repeatable improvement in verified outcomes and if explainability, appeal and correction safeguards are in place.

Recommendation 5: Monitor Evidence Quality in the Priority Queue

The evidence-tier check found a large difference in priority access at the same global 20% capacity. Random Forest selected 25.60% of complete-evidence cases and 4.89% of regional-fallback cases. The rule selected 24.67% and 7.38%, respectively. The recall comparison also changes by tier: RF captures 34.51% of losses in the complete tier against the rule's 33.18%, but only 8.23% in the fallback tier against 11.47% for the rule. RF is therefore not better for every group of contexts. This result does not establish real-world unfairness or isolate its cause. It does mean that any shadow trial should examine selection and loss capture separately by evidence tier, with particular attention to data-poor contexts.

Roadmap Stage	Management Action
Stage 1 - Shadow trial	Run the transparent rule beside current human assessment; preserve human authority, record evidence quality and monitor priority access across evidence tiers.
Stage 2 - Operational measurement	Measure assessor effort, review completion, resolution time, verified loss outcomes and correction/appeal events.
Stage 3 - Model challenge	Retrain and compare ML against the rule at the same review capacity using real verified outcomes.
Stage 4 - Controlled adoption	Expand automation only when incremental value and safeguards are demonstrated.

Risks and Limitations

The following limitations define the boundary of the current project. They are consolidated here so that the main analysis can remain decision-focused while the assumptions remain explicit.

Limitation	Implication
Synthetic supervision	Loss labels and simulator coefficients are assumptions rather than externally verified farmer outcomes. Results establish behaviour within the declared experiment, not real-world claim accuracy.
Simulator coupling	Weather, vegetation, temperature and rainfall-event variables deliberately contribute to the synthetic label mechanism. Their predictive value within the experiment is therefore partly structural and should not be interpreted as empirical discovery of PMFBY loss causality.
Granularity	Model inputs are shared at district-season-year level, and the saved-output check found identical LR/RF scores within each test context. The current system therefore prioritises contexts, not individual farmer risk.
Calibration	Spatial cutoffs, event thresholds, anomaly histories and imputation choices were not calibrated against verified crop damage.
Validation scope	The paired-context fixed-queue diagnostic does not establish a conclusive RF advantage, but it does not test new simulation seeds, model refitting, unseen districts or alternative label mechanisms.
Capacity proxy	Equal numbers of reviewed cases are treated as equal capacity, although real cases may require different assessor effort.
Operational outcomes	Settlement time, staff-hours, payment correctness, cost savings, user interface performance and feedback-loop effectiveness were not measured.
Evidence quality	Complete-evidence contexts received substantially greater priority access than regional-fallback contexts under RF and the transparent rule in the 2022 simulation. Any shadow trial should monitor this explicitly; the result is a diagnostic concern, not proof of real-world discrimination.

These limitations do not invalidate the Big Data engineering demonstration. They restrict the deployment claim: the current release is a reproducible simulation and prioritisation study that should precede, not replace, supervised real-world validation. Appendix B records the executed technical choices, their rationale and the main alternatives that would alter the experiment.

Conclusion

The final pipeline retains all 4,497 aggregate contexts and generates the two-million-record synthetic workload. It also records which model inputs came from observations and which required estimates.

Within the simulation, the weather-vegetation rule improves loss capture over random selection and performs close to Random Forest. The additional evaluation does not establish a conclusive RF advantage, and both strategies give substantially less priority to fallback-evidence cases. The next step is a shadow trial of the simpler

rule, with evidence-tier monitoring and no change to actual review decisions. A more complex model should be considered only after its additional benefit has been demonstrated.

Reproducibility and Release Record

Item	Recorded Value
Final release	v2.2.0
Executed run	r20260907_132222
Input namespace	v2.1 Gold, seasonal weather and vegetation layers at Delta version 0
Output design	Separate v2.2 run namespace with versioned Delta tables
Controls	Fixed seed 20260907; source hashes; ten passing acceptance checks
Acceptance scope	Retained contexts, exact unique volume, complete model inputs, temporal separation, scoring coverage and equal-capacity comparison
Saved-output evaluation	The supplement evaluated the saved v2.2 outputs without generating new scenarios or refitting models. All 18 policy/capacity totals matched the original results. The source acceptance checks still passed, as did three local tests of the numerical checks used for context resampling.

References and Source Register

1. Pradhan Mantri Fasal Bima Yojana (PMFBY), Ministry of Agriculture & Farmers Welfare, Government of India - <https://pmfby.gov.in/>
2. NASA POWER Daily API Documentation - <https://power.larc.nasa.gov/docs/services/api/temporal/daily/>
3. MODIS/061/MOD13Q1 Terra Vegetation Indices 16-Day Global 250m, Google Earth Engine Data Catalog / NASA LP DAAC - https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD13Q1
4. Apache Spark MLlib Classification and Regression Documentation - <https://spark.apache.org/docs/latest/ml-classification-regression.html>

AI Usage Disclosure

OpenAI ChatGPT/Codex assisted with code development and debugging. The submitted team remains responsible for reviewing the assumptions, confirming the final methodological choices and accepting responsibility for the interpretation presented in the final submission.

Team Contributions

Roll No.	Member	Actual Contribution
C004	Navneet Narula	Engineering and pipeline: ingestion, joins, missing-data handling, versioning, reproducibility and execution demonstration.
C010	Keshav Arora	Policy evaluation: equal-capacity comparisons, baseline strategies, prioritisation worklist and business interpretation.
C030	Piyush Joshi	Data foundation: source descriptions, data dictionary, lineage, coverage and Big Data characteristics.
C036	Vedanti Agrawal	Modelling: synthetic-generation explanation, model inputs, modelling choices and leakage safeguards.
C043	Bhadrashatte Manav	Validation: reconcile reported metrics with outputs, explain the expected probability diagnostic, uncertainty and evidence-tier differences.
C051	Aman Agrawal	Business and deployment: assessor workflow, safeguards, limitations, recommendations and presentation integration.

Appendix A: Key Simulation Settings

Setting	Executed Definition
Scenario volume	Exactly 2,000,000 records; each real context receives at least one scenario, with the remainder allocated 50% uniformly and 50% by insured exposure.
Base probability	7% simulated loss probability before stress additions.
Rainfall stress	+22 percentage points at rainfall anomaly $\leq -30\%$; otherwise +10 points at $\leq -10\%$.

Setting	Executed Definition
Vegetation stress	+20 percentage points at NDVI anomaly ≤ -0.15 ; otherwise +8 points at ≤ -0.05 .
Event additions	+8 points when maximum daily-mean temperature ≥ 30 C; +5 points when at least three days have rainfall ≥ 20 mm.
Randomness and bounds	Independent uniform shock up to 0.12; final probability bounded to 0.02-0.95 before the binary event draw.
Transparent rule	One point each for rainfall anomaly $\leq -10\%$, NDVI anomaly ≤ -0.05 and maximum daily-mean temperature ≥ 30 C; compared at 10%, 20% and 30% review capacity.

Appendix B: Technical Decision Register

This register preserves the detailed implementation logic behind the final experiment without interrupting the main consulting narrative. It records what was executed, why the choice was made, and how a plausible alternative could change coverage, ranking, runtime or interpretation.

B1 Spatial distance and acceptance

Execution: Computed Haversine distance assuming an Earth radius of 6,371.0088 km, with nearest available NASA request location rounded to nearest within 75 km.

Rationale: The Earth radius is used to convert the distance into km. There was no attempt to tune the 75 km limit to the crop-loss outcome, it was simply an engineering decision to keep spatial mismatch out of the results and not to eliminate matches altogether.

Sensitivity: At 50 km a slightly smaller group of matches would have been accepted; 100 km: a more distant group of matches would have been accepted. Both were not tested - thus, no measurement of the resulting change in coverage or accuracy was performed.

B2 Input validity checks

Execution: Coordinates kept within the range latitude -90 to 90, and longitude -180 to 180. Screen daily temperature to -90 to 65 C, rainfall to 0-3,000 mm and humidity to 0-100%. The code checks for invalid dates, coordinates and measurements and removes/marks them as missing as appropriate.

Rationale: This is a valid representation check for the coordinate and percentage limits. The broad ranges of temperature and rainfall data are simple data screens. They were not chosen as cut out of yield damage.

Sensitivity: If the sensitivity is tightened, more observations would be flagged and fewer days would be available for use. No comparisons with alternative screening limits were made, and thus the impact of the limits on seasonal coverage is not quantified.

B3 Season calendars and completeness

Execution: Kharif – June to October, Rabi – November to April; also, Rabi is assigned to its starting year. Leap days are included when calculating the expected number of days in each season.

Rationale: The seasonal joins are consistent when using a common calendar. The 95% gate is a data quality setting that was selected; it has not been determined that this is best cut-off or the same calendar is best for all crops and districts.

Sensitivity: The event counts and available evidence could be different if there is sensitivity / alternative information, such as local crop calendars or a 90% or 100% gate. Those alternatives were not tested and their impact on the outcomes is not measured.

B4 Dry days heat days and rainfall events

Execution: Count dry days with rainfall below 1 mm, heat days with daily-mean temperature at least 35°C, and rainfall days meeting thresholds of at least 20 mm and at least 50 mm separately.

Rationale: These are summaries that add daily weather information into the seasonal model inputs. Thresholds are not related to any damage validation, and are to be interpreted as feature definitions not validated damage triggers.

Sensitivity: The features would change if other thresholds / time windows were used. They were not tested for their effects on ranking. Evidence that would not be developed in this experiment would be needed to set crop-specific thresholds.

B5 Prior year anomaly construction

Execution: Calculate anomalies using earlier years only, requiring at least two prior years. Exclude the current year and future years from the historical baseline. Rainfall anomaly is a percentage departure; NDVI anomaly is an absolute departure.

Rationale: minimum of 2 years avoids single observation. This is a minimum practical rather than a long term climate baseline or setting validated against a crop loss.

Sensitivity: If a minimum of one year is used, then more anomalies would be available. Longer histories would give more contexts a longer history, but would also cause more contexts to be without one. These were not compared to the conventional varieties.

B6 Scenario count and allocation

Execution: Generate exactly 2,000,000 synthetic scenarios. Allocate at least one scenario to every context, then allocate the remaining scenarios using a 50% uniform and 50% exposure-weighted split, with integer allocation preserving the exact total.

Sensitivity: If the remaining rows were to be distributed evenly using a uniform-only allocation, it would be more uniform. An exposure-only allocation would weight the high-exposure contexts more highly. Either would modify aggregate statistics, but both options were not tested.

B7 IDs seed and synthetic amount

Execution: Use version, context, sequence and seed 20260907. The amount, the shock, the event and the sampling draws are each provided by a separate deterministic hash stream. Calculate the simulated amount as $(\text{context insured exposure} \div \text{allocated scenario count}) \times (0.75 + 0.5U)$, where U is a uniform draw between 0 and 1.

Rationale: The generated records are reproducible because they are fixed (not randomized) and are separate from each other. The amount multiplier includes a range of assumptions and is not a fitted distribution to the observed amounts of farmers.

Sensitivity: Individual outcomes/exposure rankings may change if other seeds or amount distributions were used. Comparisons were not made of repeated seed or distribution.

B8 Base probability and rainfall additions

Execution: The simulation begins with a 7% loss probability. It adds 22 percentage points when the rainfall anomaly is at most -30%. Otherwise, it adds 10 percentage points when the rainfall anomaly is at most -10%.

Rationale: The formula is based on the assumption that there is an increase in loss risk as rainfall deficit increases. The starting rate, cutoffs and additions were selected for the simulation and not derived from the outputs of PMFBY losses. No empirical values for these were set.

Sensitivity: If these values are changed, the loss prevalence will also change and the relationship learned by the models will change. The project did not attempt to compare alternative settings, therefore their influence on performance is not measured.

B9 Vegetation additions

Execution: If NDVI anomaly is at most -0.15, add 20 percentage points to loss probability. Otherwise, if it is at most -0.05, add 8 percentage points. These vegetation additions are mutually exclusive.

Rationale: This is because a negative vegetation deviation will add to simulated loss risk. An additional source of evidence is vegetation which has not been shown to be statistically independent of weather. The additions and thresholds are still assumptions.

Sensitivity: The resulting vegetation labels and the importance of vegetation in the model would vary if it were not a continuous relationship or if the cutoffs were altered. These alternatives were not proven or adjusted for actual loss.

B10 Temperature and heavy-rain additions

Execution: Add 8 percentage points to loss probability when the maximum daily-mean temperature is at least 30°C. Independently, add 5 percentage points when at least three days have rainfall of at least 20 mm. Both contributions may apply.

Rationale: The simulation assumes that elevated temperature and repeated heavy rainfall can contribute additional loss risk. The 30°C threshold is used in the simulated probability formula, while the 35°C threshold defines the separate heat-day feature. Neither threshold was calibrated as a crop-specific damage limit.

Sensitivity: If other temperature limits, rain-day counts or interactions change, target will change. They were not quantified as there was no sensitivity comparison performed.

B11 Hidden shock probability bounds and binary event

Execution: The simulation adds an unobserved uniform random shock between 0 and 0.12 to the loss probability. The resulting probability is then restricted to the range 0.02 to 0.95. A separate uniform random draw determines whether the scenario receives label 1 for simulated loss or label 0 for simulated non-loss.

Rationale: Shock is added as it provides variation that the model is not able to observe. Even if the event's probabilities are known, the separate event draw makes the event result a random outcome. The bounds are used to prevent the probability from reaching certainty, but the exact value of the bounds were not calibrated from the empirical data.

Sensitivity: A change in the shock or the rule for generating the events would alter the simulation. The shock removal would not make the labels deterministic, as there would still be an event draw. The effect of alternative settings on AUC is not measured, as they were not tested.

B12 Classification thresholds

Execution: Positive-class F1 with search thresholds between 0.05 and 0.65, with a step size of 0.05, was executed. The value of 0.05 was selected for Logistic Regression and 0.20 for both versions of Random Forest.

Rationale: The coarse grid restricts the search, and the additional validation year ensures that the use of thresholds does not influence the end results of the test. This selection didn't yield any useful test set classification: all cases were classified as positive.

Sensitivity: Different thresholds, calibration methods or capacity-based cutoffs might change the class decisions. No comparison between them was made. The end-of-queue comparison is done using continuous ranks and the non-successful classification result is reported.

B13 Transparent rule score and review capacity

Execution: Assign one point each for rainfall anomaly at most -10% , NDVI anomaly at most -0.05 , and maximum daily-mean temperature at least 30°C . Rank by the total rule score, using the documented identifier-based tie-break.

Rationale: It is easier to check the rule if the weights are equal. The strategies are comparable within the experiment since they have the same queue sizes. The weights, thresholds and capacities were not designed to fit assessor workloads.

Sensitivity: depending on different weights or a budget in terms of staff-hours, the cases that get into the queue might vary. These were not tested and the project does not make any claims as to the operational optimality of the settings.

B14 Review routes

Execution: The top 20% form the priority-review batch. Remaining cases stay in the human-review workflow. Priority determines review order and does not automatically approve or reject a claim.

Rationale: Human assessment is needed for all of the routes. The routing thresholds are design assumptions, NOT proven services standards. It was reported an empty fast-review route and it was not filled by switching the cutoffs after viewing the outcome.

Sensitivity: Other cutoffs would have resulted in putting cases in that route, but were not tested. There is no evidence in the current experiment that such cases would be processed more quickly or correctly.

B15 Expected probability diagnostic

Execution: Independent 0-0.12 shock: For the diagnostic, use its mean contribution of 0.06. Make comparisons between the resulting expected probability and the sampled labels. This diagnostic score should not be used in the Policy Competition.

Rationale: To examine the amount of ranking information available with the selected simulator and to understand the model AUC close to 0.57. Does not check the assumed loss mechanism against real claims.

Sensitivity: Different simulator would have different benchmark. The diagnostic is therefore specific to the experiment and not necessarily a performance ceiling or a metric for the accuracy of the claims in the real world.